

Vertical Elaboration or Horizontal Integration? A Citation-Network Analysis of Saudi Arabian Legislation

Abstract

Network analysis of legal citation has produced a substantial literature for the United States and Europe, but the legal systems of the Arab world have remained outside it, for want of machine-readable statutory corpora. Using a newly released, provenance-annotated corpus of Saudi Arabian legislation, we construct and analyse what is, to our knowledge, the first citation network over Saudi statutory law: 290 legislative instruments, 15,689 articles, and 585 extracted inter-instrument references. We first address a problem that the citation literature typically leaves implicit: the extractor that resolves a cited law name to a target instrument is itself fallible. Validating every resolved edge against the target’s registered title and adjudicating the residue by hand, we find a 3.3% misresolution rate and analyse only the 410 surviving citations. We then separate *vertical* citations, in which a subordinate instrument elaborates its own parent statute, from *horizontal* citations that cross instrument families. The distinction is not cosmetic: the Labour Law appears to rival the Companies Law as the system’s leading hub until its eight annexes are recognised as part of its own family, after which its horizontal citation count falls by more than a third and the Companies Law stands alone. Horizontal citations nevertheless account for 70.5–74.4% of the network depending on how family boundaries are drawn, so Saudi legislation is genuinely cross-referential rather than merely hierarchical. The horizontal network is sparse (164 instruments, 206 edges) and highly concentrated: ten instruments receive 58% of all horizontal citations, led by the Companies Law (46 citations from 26 distinct instruments). Breadth and depth diverge sharply — the Capital Market Law draws 26 citations from only 9 instruments, while the Criminal Procedure Law draws 13 from 13 — a contrast that raw citation counts conceal. Cross-referencing the citation network against a hand-classified supersession graph, we identify live instruments that still cite repealed predecessors, and we map the instruments that Saudi legislation cites but that no consolidated machine-readable source yet covers. We discuss what these structures imply for legal informatics, retrieval-augmented legal AI, and legislative maintenance.

Keywords: Legal citation networks · Saudi Arabia · Legislation · Legal informatics · Network analysis · Legal data quality

1 Introduction

A legal system is not a list of laws. Statutes refer to one another: they borrow definitions, defer to procedures, carve out exceptions, and assign jurisdiction. Those references form a network, and for two decades that network has been an object of quantitative study — initially for judicial precedent

(Fowler et al., 2007), then for statutory corpora (Katz and Bommarito, 2014; Coupette et al., 2021). The results have been substantive rather than decorative: they identify which instruments carry structural load, how complexity accumulates, and how legal systems change over time.

That literature is almost entirely Anglo-American and European. The legal systems of the Arab world are absent from it, and the reason is prosaic rather than intellectual: the analysis requires a machine-readable statutory corpus with resolvable citations, and for those jurisdictions no such resource has been available. Saudi Arabia is a case in point. Its legislation is published officially in the *Umm Al-Qura* gazette and on government portals as human-readable documents; it has recently undergone a major codification wave, including a Civil Transactions Law of 721 articles, a new Companies Law, a new Evidence Law, and a new Personal Status Law; and nothing about its citation structure has been measured.

This paper reports what is, to our knowledge, the first such measurement. It uses a recently released corpus of Saudi legislation — 290 instruments and 15,689 article-level records with per-record provenance (Author(s) withheld for double-anonymous review, 2026) — and the citation graph derived from it, and asks three questions.

Is the network hierarchical or integrated? Saudi legislation, like most civil-law systems, is organised in families: a statute and the implementing regulation that elaborates it. If most citations simply run from a regulation to its own parent, the network is an artefact of drafting hierarchy and says little about systemic integration. We therefore separate vertical from horizontal citation and analyse them apart (Section 3.3). They behave differently, and the separation changes which instruments count as central.

Which instruments carry structural load? We report centrality over the validated horizontal network and distinguish instruments cited by many others from instruments cited often by few (Section 4.2).

Does the network contain maintenance defects? Cross-referencing the citation graph against a hand-classified supersession graph, we look for live instruments that still cite repealed predecessors (Section 4.4), and we map what Saudi legislation cites that is not yet available in consolidated machine-readable form (Section 4.5).

A fourth contribution is methodological and cuts across the rest. Citation extraction over statutory text involves two distinct steps — detecting that a citation occurred, and resolving the cited name to an instrument — and the second is fallible in ways that quietly corrupt network results. We validate every resolved edge and report the error rate rather than assuming it away (Section 3.2). We recommend this as standard practice for legal citation-network studies built on automatic extraction.

2 Related work

Legal citation networks. Fowler et al. (2007) established the network-analytic study of legal citation with an analysis of United States Supreme Court precedent, using centrality to measure the legal importance of individual cases. Katz and Bommarito (2014) extended the approach from case law to statutory law, treating the United States Code as a structured object whose complexity can be measured. Coupette et al. (2021) generalised the framework to legislation over time and across

jurisdictions, comparing the statutory and regulatory corpora of the United States and Germany. This paper follows that statutory line rather than the precedential one: the unit of analysis is an enacted instrument, and the edges are cross-references between instruments.

Legal corpora and standards. The machine-readable substrate for such work has grown considerably — Akoma Ntoso (Palmirani and Vitali, 2011) and the European Legislation Identifier for structure, and large corpora such as the Pile of Law (Henderson et al., 2022) and MultiLegalPile (Niklaus et al., 2024) for text. None covers Saudi legislation, and Arabic is at best marginal in them. The corpus used here (Author(s) withheld for double-anonymous review, 2026) was built specifically to close that gap; the present paper is an analysis of the legal system it describes rather than a description of the resource.

Citation extraction and its reliability. Work on legal citation extraction has generally reported detection performance, while the downstream step — resolving a detected citation to the instrument it names — receives less scrutiny in network studies that consume its output. Where corpora are curated by hand this matters little. Where edges are produced by automatic resolution, as here, misresolution propagates directly into centrality rankings, and we treat it as a first-class methodological concern (Section 3.2).

3 Data and method

3.1 Corpus and citation graph

The corpus comprises 290 Saudi legislative instruments — 201 statutory laws, 64 statutory regulations, 24 implementing regulations, and one set of procedural rules — structured into 15,689 article-level records with per-record source provenance under a four-tier verification taxonomy (Author(s) withheld for double-anonymous review, 2026). A deterministic extractor scans every record for statutory citations and emits 3,607 references, of which 3,022 are internal to a single instrument and 585 point from one instrument to another. This paper analyses the 585 inter-instrument references; internal cross-references describe the architecture of individual statutes rather than the structure of the system.

Of the 585, the extractor resolves 424 to a specific instrument in the corpus. The remaining 161 name an instrument the corpus does not contain; they are excluded from the network but analysed separately in Section 4.5, where they turn out to be informative in their own right.

3.2 Edge validation

Resolution errors are not hypothetical. The extractor matches a cited name — often truncated, inflected, or embedded in surrounding text — against the corpus’s registered instrument titles, and some matches are simply wrong. In one case a citation to the (uncodified) civil-service discipline regime resolves to the Contractors Classification Law; in another, a reference to the phrase “Islamic Sharia” resolves to the Public Prosecution Law. Left in place, such edges add spurious in-degree to arbitrary instruments.

We therefore validated every resolved edge before analysis. Each cited name and the registered Arabic title of its resolved target were normalized (diacritics and tatweel stripped, alef, teh-marbuta and alef-maqsurat variants folded) and reduced to content tokens by removing structural words such as “law”, “regulation”, “implementing”, “previous”, and decree formulae. An edge whose cited name shares at least one content token with its target’s title was accepted automatically; 408 of 424 edges cleared this test. The remaining 16 were adjudicated by hand against the raw citation text, and the decisions are published with the analysis code. Two proved correct — a morphological variant of “bankruptcy”, and a truncation of “valuers” — and 14 were misresolutions and were dropped.

The resulting misresolution rate among resolved edges is **3.3%** (14/424). Analysis proceeds on the surviving **410** citations. We report this figure not as a criticism of the extractor, whose published caveats already flag confidence levels, but because a network study that consumes automatic resolution silently inherits its errors, and 3.3% is large enough to move a ranking of instruments whose citation counts differ by one or two.

3.3 Vertical and horizontal citation

Saudi legislation is drafted in families: a statute is typically accompanied by an implementing regulation, and sometimes by annexes, model forms, and procedural rules. A regulation necessarily cites its own parent statute, often repeatedly. Counting those citations alongside references that cross from one body of law into another conflates two different phenomena.

Family membership must therefore be recovered before anything is counted, and how it is recovered matters. Deriving families by stripping structural suffixes from instrument identifiers is the obvious approach and it is wrong: it splits the Labour Law’s model contract forms and accessibility arrangements from the Labour Law itself, and would classify 36 parent-child citations as horizontal. We instead take family membership from the corpus’s own subject-matter grouping, which joins each statute with its regulation and annexes — the Labour Law family spans eight components — and recover it by mapping every track’s data files to the grouping key its records carry. A residual ambiguity remains where a subordinate instrument carries its own grouping key derived from the parent’s (bankruptcy fees under bankruptcy), so we report both readings: a *strict* one that treats such keys as separate families and a *prefix* one that merges them, giving upper and lower bounds on the horizontal share.

The distinction is consequential in two directions. Instruments that appear prominent on raw counts — the Anti-Narcotics Law (15 citations) and the Traffic Law (10) — receive those citations from exactly one source each, their own implementing regulation, and disappear from the horizontal ranking entirely. More strikingly, the Labour Law falls from 36 horizontal citations under the naive family definition to 23 under the corrected one, because thirteen of its citations come from its own annexes. Under the naive definition the Labour Law and the Companies Law look like twin hubs of comparable weight; under the corrected one the Companies Law stands alone. An analysis that skipped this step would have reported a headline finding that the data do not support. Unless stated otherwise, the results below concern the horizontal network under the prefix reading.

Measure	Value
Instruments in the corpus (nodes)	290
Instruments with at least one citation	201
Distinct citing pairs (edges)	257
Citations carried	410
Vertical citations, strict family reading	105
Vertical citations, prefix family reading	121
Horizontal citations (across families)	289–305
Horizontal share	70.5–74.4%
Horizontal network: instruments	164
Horizontal network: edges	206
Share of horizontal citations to the top ten	58.1%
Density over the citing subgraph	0.0064
Weakly connected components	12
Largest component	175
Reciprocal instrument pairs	2

Table 1: Structure of the validated Saudi legislative citation network.

4 Results

4.1 Global structure

Table 1 summarises the network. Three features stand out.

First, the network is **predominantly horizontal**: between 289 and 305 of 410 citations — 70.5% to 74.4% depending on where family boundaries are drawn — cross instrument families. Saudi legislation is therefore genuinely cross-referential; the citation structure is not an artefact of statute-plus-regulation drafting. This is worth stating plainly because the opposite result was entirely plausible *a priori* for a codified civil-law system with a strong statute/regulation hierarchy. It is equally worth noting that a naive family definition inflates this figure to 79.3%, which is why we report the bounds rather than a single number.

Second, the network is **sparse but connected**. Density over the citing subgraph is 0.0064, yet 175 of the 201 citing instruments fall into a single weakly connected component. Saudi law is not a set of isolated regimes: nearly every instrument that cites anything is, directly or transitively, connected to nearly every other. The remaining 11 components are small and topically isolated.

Third, citation is **almost entirely asymmetric**: only two pairs of instruments cite each other. Legislative cross-reference in this corpus runs one way, from later or subordinate instruments toward a small set of established ones — the structure of a system with a settled core rather than one in negotiation with itself.

4.2 Structural hubs

Figure 1 shows the horizontal in-degree of the leading instruments, with the number of distinct citing instruments alongside.

The **Companies Law** is the system’s dominant hub: 46 horizontal citations from 26 distinct instruments, nearly twice the count of any other instrument and drawn from the widest range of sources. Its prominence is intelligible — nearly every economic regulation must settle who counts

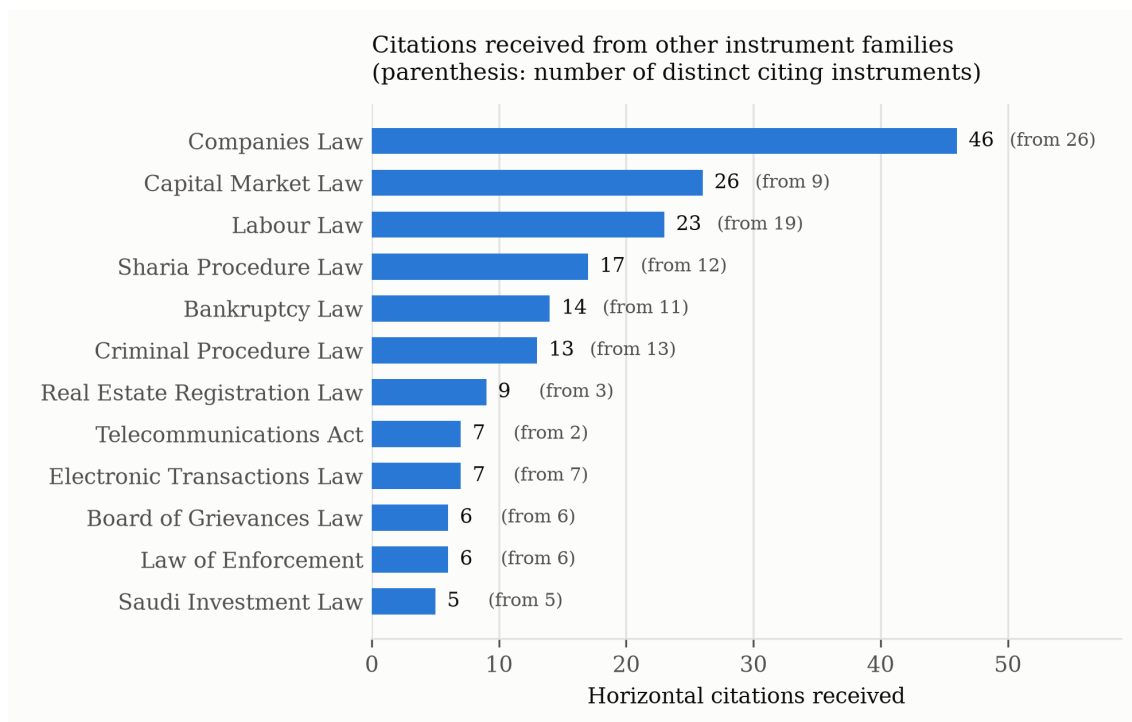


Figure 1: Horizontal citations received by the twelve most-cited instruments. The figure in parentheses is the number of *distinct* instruments that cite the target, which separates breadth from depth: the Capital Market Law receives 26 citations but from only 9 instruments, while the Criminal Procedure Law receives 13 citations from 13 distinct instruments.

as a company — and it is robust to how family boundaries are drawn. The **Labour Law** follows at 23 citations from 19 instruments, still the second-broadest reach in the corpus but no longer a rival to the Companies Law once its own annexes are excluded. That commercial and labour statutes rather than, say, constitutional instruments occupy these positions tells us something real about which parts of Saudi law are load-bearing in day-to-day regulation.

Citation is heavily concentrated: the ten most-cited instruments receive 58.1% of all horizontal citations, while 164 instruments participate in the horizontal network at all. A small definitional core carries the system.

Breadth and depth come apart sharply. The **Capital Market Law** ranks second by citation count (26) but draws them from only nine instruments: it is cited intensively within the financial regulatory cluster and barely outside it. By contrast the **Criminal Procedure Law** receives 13 citations from 13 distinct instruments, and the **Electronic Transactions Law** 7 from 7 — perfectly dispersed citation, the signature of an instrument that supplies a general mechanism (here, the legal status of electronic records) that many unrelated regimes need exactly once. A ranking by raw count would treat the Capital Market Law as more central than either; a ranking by breadth would reverse it. Both readings are legitimate and they answer different questions, which is precisely why the two figures should be reported together.

Weighted PageRank over the validated network puts the Companies Law first (0.075), the Capital Market Law second (0.066), the Labour Law third (0.047), and the Bankruptcy Law fourth (0.036). PageRank rewards the Capital Market Law for being cited by instruments that are themselves cited, which is consistent with its position at the centre of a dense financial cluster.

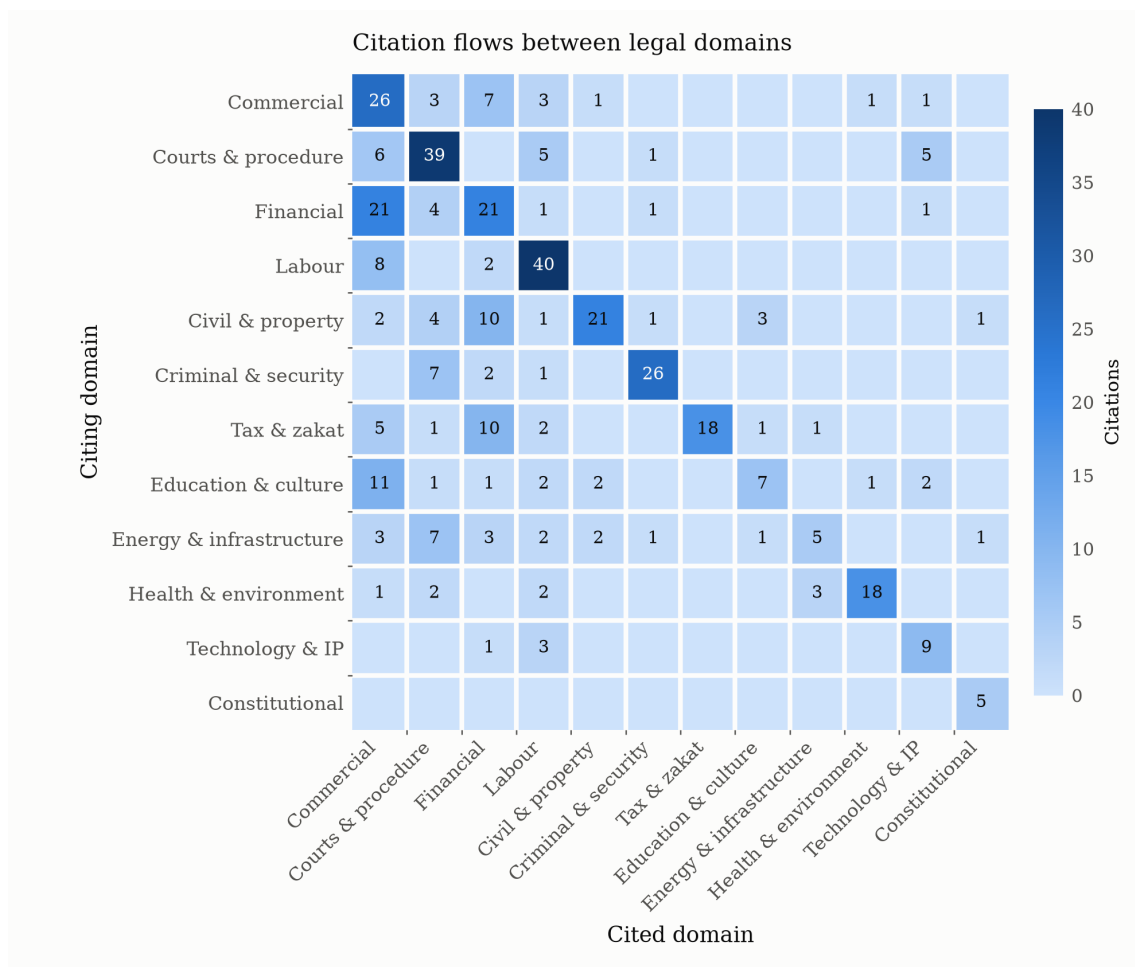


Figure 2: Citation flows between legal domains. The diagonal carries intra-domain citation; off-diagonal cells show which bodies of law reach into which others. Domains follow the author-assigned grouping released with the corpus.

As a robustness check we recomputed the horizontal share and the leading hubs on all 424 resolved edges, that is without dropping the 14 misresolutions. The horizontal share moves only from 70.5% to 71.5% and the four leading hubs are unchanged, so the headline structural findings do not depend on the validation step. The step still matters for individual rankings further down the distribution, where instruments separated by one or two citations change places — which is precisely where a reader interested in a specific instrument would look.

4.3 Domain structure

Aggregating citations to the domain level (Figure 2), 235 of 410 citations stay within a domain and 175 cross domains. The diagonal dominates, as expected, but the off-diagonal structure is informative.

The heaviest cross-domain flow runs from **financial and capital markets into commercial and corporate law** (21 citations) — financial regulation resting on company-law foundations. Tax and zakat instruments cite financial ones (10); civil and property instruments cite financial ones (10); education, culture and civil-society instruments cite commercial law (11), reflecting that associations, cooperatives, and professional bodies are constituted through commercial-law

machinery. Courts and procedure sends citations outward (to commercial law, 6; to technology and IP, 5) more than it receives from most domains, which is what one expects of procedural law: it is invoked by substantive regimes rather than invoking them.

The constitutional and administrative domain is almost entirely disconnected from the rest of the citation network. This is not evidence that constitutional instruments are unimportant; it reflects that foundational instruments are presupposed rather than cited, a pattern also observed in statutory networks elsewhere.

4.4 Dangling citations to repealed instruments

The corpus ships a supersession graph of 102 repeal and replacement relationships, each classified by hand from the instruments' own documented text. Cross-referencing it against citations whose target the extractor could *not* resolve — on the hypothesis that an unresolvable cited name may be a repealed instrument no longer in force — identifies citations that point at law that has been superseded.

Matching is deliberately conservative. A cited name matches a repealed instrument only when all of its content tokens appear in that instrument's recorded description and it carries at least two such tokens. The two-token floor is essential: single generic tokens such as *al-hay'ah* ("the Authority") or *al-'amal* ("work") are contained in many descriptions and generate false matches. Applying it reduces an initial seven candidates to **three confirmed cases**:

- The *Implementing Regulation for Environmental Inspection and Audit* (Article 8) and the *Implementing Regulation for Environmental Service Providers* (Article 12) both direct the reader to the implementing regulation on recording environmental violations and imposing penalties — in the version that has since been replaced.
- The *Foreigners' Residency Law* (Article 56) provides for prosecution under the General Civil Servants Law (*niẓām al-muwazzafīn al-'āmm*), an instrument repealed in full by the Civil Service Law.

Three cases in a corpus of this size is a small number, and we resist inflating it: the true figure is bounded below by our method, since we can only detect repeals that the supersession graph records and citations whose names are distinctive enough to match. What the finding demonstrates is that the defect class *exists* and is now *detectable at scale*. In a jurisdiction without a machine-readable consolidated gazette, a citation pointing at repealed law is discoverable only by a reader who happens to know the repeal — or, from now on, by intersecting two released graphs.

4.5 What Saudi legislation cites but no corpus contains

The 161 references the extractor could not resolve are not merely extraction noise. They name **128 distinct** instruments and provisions that Saudi legislation actively relies on and that the corpus — the most comprehensive machine-readable collection of Saudi law available — does not contain. Some are older instruments superseded but still cited (Section 4.4); others are live instruments not yet consolidated in machine-readable form, among them retirement and pension regimes, social-care

regulations, and internal committee rules; and a residue are formulaic references to instruments issued by decision of an authority, which name no specific instrument at all.

Read as a coverage map, this set is a prioritised work list for anyone extending machine-readable coverage of Saudi law: it identifies exactly which absent instruments the existing corpus most often needs.

5 Discussion

For legal informatics. The vertical/horizontal distinction generalises beyond Saudi Arabia. Any jurisdiction that pairs statutes with implementing regulations will produce citation networks in which hierarchical elaboration and systemic integration are mixed, and any centrality ranking computed over the mixture will overstate the importance of statutes that happen to carry long regulations. Reporting the split costs little and changes conclusions.

For retrieval-augmented legal AI. Systems that answer questions about Saudi law by retrieving statutory text inherit the structure measured here. Two implications follow. Retrieval that returns a single article will frequently be insufficient, because the governing rule lives in another instrument that the retrieved article cites — most often the Companies Law or the Labour Law, which between them receive 69 horizontal citations from 45 distinct instruments. And a system that grounds answers in statutory text without checking supersession can reproduce dangling citations of exactly the kind identified in Section 4.4, faithfully quoting a provision that points at repealed law.

For legislative maintenance. The graphs released with this analysis make two maintenance questions computable: which live instruments cite superseded law, and which cited instruments lack a consolidated machine-readable text. Neither is answerable by reading, at this scale, by anyone.

6 Limitations

We measure the extractor’s output, not the true citation network. The 585 inter-instrument references are those a deterministic pattern-based extractor detected. Its recall is unmeasured: citations phrased in ways the patterns do not cover are invisible to this study, and the resulting network is therefore a lower bound on the system’s true connectivity. Precision, by contrast, we do measure, and correct, for the resolution step (Section 3.2).

Coverage bounds the network. The corpus covers 290 instruments, not every Saudi legislative instrument. Citations to absent instruments become unresolved edges rather than network structure, so measured centrality is centrality *within the corpus*.

Dangling-citation detection is conservative. It depends on the supersession graph, which is hand-classified and deliberately incomplete, and on name matching, which we constrained to avoid false positives at the cost of recall. Three confirmed cases is a floor, not an estimate.

Family assignment is heuristic. Vertical and horizontal citation are separated using the corpus’s subject-matter grouping, which is a good but imperfect proxy for legal subordination. A subordinate instrument issued under a differently named parent — a capital-market authority regulation issued under the Capital Market Law, for instance — is counted as horizontal. The two readings we report bound the effect of the ambiguity we could detect; they do not bound this one, which would require explicit parent-instrument metadata that no current source provides.

Hub interpretation is bounded by coverage and by counting. Instruments outside the corpus cannot accumulate citations, and citation counts measure how often an instrument is invoked, not how consequential each invocation is: a single reference that imports an entire licensing regime counts the same as a passing definitional cross-reference. Centrality here is a measure of referential load, not of legal importance.

The domain grouping is author-assigned. Domain membership is not a property of Saudi legislation; the twelve-domain grouping used in Figure 2 is applied by published rules and is an organising device, not a legal taxonomy.

The network is a snapshot. All instruments are analysed as of the corpus release. Because the corpus records publication dates, the temporal analysis that Coupette et al. (2021) perform for other jurisdictions is possible in principle, and we leave it to future work.

7 Conclusion

We have constructed and analysed what is, to our knowledge, the first citation network over Saudi Arabian legislation: 290 instruments, 410 validated inter-instrument citations, and a 3.3% resolution-error rate measured rather than assumed. The network is 70.5–74.4% horizontal, sparse, overwhelmingly asymmetric, and concentrated in a single component of 175 instruments centred on the Companies Law, with the ten most-cited instruments absorbing 58% of horizontal citation. Separating vertical from horizontal citation was not a refinement but a correction: under a naive family definition the Labour Law appears to co-lead the system, and it does not. Separating breadth from depth then reveals two further kinds of importance — the intensively cited Capital Market Law against the widely cited Criminal Procedure and Electronic Transactions Laws — that a raw citation count merges. Intersecting the citation network with a supersession graph makes a previously invisible defect class detectable: live instruments that cite repealed law.

The wider point is that this analysis became possible the moment a provenance-annotated statutory corpus existed. The same is true of every jurisdiction currently absent from the legal-network literature, which is to say most of them.

Data availability

The corpus analysed here, its citation and supersession graphs, and the analysis and figure scripts that reproduce every number and figure reported in this paper are openly available under the MIT licence and archived in a public repository with a citable DOI. The repository and DOI are withheld

here because they identify the author, and will be supplied to the editors on request and stated in full in the published version.

Legal status. This paper analyses a non-official research corpus. The corpus is not an official government publication, contains no official translation, and nothing here constitutes legal advice; the binding text of any instrument is the Arabic original published in *Umm Al-Qura*.

References

Author(s) withheld for double-anonymous review. A provenance-annotated, article-level corpus of Saudi Arabian legislation, 2026. Manuscript under review. The corpus is openly archived with a citable DOI; the reference is withheld here because it identifies the author and will be supplied on request.

Corinna Coupette, Janis Beckedorf, Dirk Hartung, Michael Bommarito, and Daniel Martin Katz. Measuring law over time: A network analytical framework with an application to statutes and regulations in the United States and Germany. *Frontiers in Physics*, 9, 2021.

James H. Fowler, Timothy R. Johnson, James F. Spriggs, Sangick Jeon, and Paul J. Wahlbeck. Network analysis and the law: Measuring the legal importance of precedents at the U.S. Supreme Court. *Political Analysis*, 15(3):324–346, 2007.

Peter Henderson, Mark S. Krass, Lucia Zheng, Neel Guha, Christopher D. Manning, Dan Jurafsky, and Daniel E. Ho. Pile of Law: Learning responsible data filtering from the law and a 256GB open-source legal dataset. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022), Datasets and Benchmarks Track*, 2022.

Daniel Martin Katz and Michael J. Bommarito. Measuring the complexity of the law: The United States Code. *Artificial Intelligence and Law*, 22(4):337–374, 2014.

Joel Niklaus, Veton Matoshi, Matthias Stürmer, Ilias Chalkidis, and Daniel E. Ho. MultiLegalPile: A 689GB multilingual legal corpus. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics*, Bangkok, Thailand, 2024. Association for Computational Linguistics.

Monica Palmirani and Fabio Vitali. Akoma-Ntoso for legal documents. In *Legislative XML for the Semantic Web*, pages 75–100. Springer, 2011.